



Mapping artificial intelligence bias: a network-based framework for analysis and mitigation

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Abstract

In the evolving era of generative artificial intelligence (AI), bias in AI systems remains a significant concern, leading to unfair outcomes across applications. Addressing this issue requires understanding how biases are introduced and propagated through AI models. In this paper, we present a network-based framework to map, analyze, and mitigate biases in AI systems. Our approach includes a systematic review of literature to identify key sources of bias, categorized into three areas: Data, Design, and human-AI teaming. Using a network-based methodology, we illustrate the interconnected nature of bias propagation in AI systems. This framework highlights critical points where bias can be introduced and offers strategies for mitigating them. We demonstrate its effectiveness through use cases in healthcare and recommender systems, underscoring the importance of integrating ethical considerations throughout the AI development lifecycle and the necessity of continuous bias monitoring. The framework provides actionable insights for developers, practitioners, and decision-makers, guiding them toward more equitable AI systems. Our paper advocates for a holistic, collaborative approach to mitigating bias, contributing to inclusive technological ecosystems.

Keywords AI bias · Algorithmic bias · Bias mitigation · Data bias · Human-AI interaction · Fairness in AI · Generative AI · Recommender systems

1 Introduction

The advent and rapid evolution of artificial intelligence (AI) in the last few years, especially in generative AI and foundational models, have ushered in transformative changes across various sectors, marking a significant shift in how enterprises approach technology and innovation. As of 2024, the global AI market is valued at approximately \$515.31 billion and is projected to surge to \$2,740.46 billion

by 2032, reflecting a compound annual growth rate (CAGR) of 20.4% from 2024 to 2032 [1]. Additionally, the AI market is expected to grow at a CAGR of 36.6% from 2024 to 2030, reaching \$1,811.75 billion by 2030 [2]. These figures highlight the substantial growth and investment in AI technologies, driven by advancements in generative AI and foundational models. This remarkable growth trajectory underscores AI's burgeoning role in reshaping industries, from healthcare, where around half of US-based providers are integrating AI tools for enhanced patient care and operational efficiency [3], to finance, where 91% of US banks employ AI in fraud detection [4]. Furthermore, the anticipated AI-driven growth of global Gross Domestic Product (GDP) by \$15.7 trillion by 2030 illustrates the scale of economic impact AI is poised to deliver [5]. Yet, 34% of companies globally have deployed AI in their systems, while 42% are still exploring Artificial Intelligence [6].

The application of AI in various sectors highlights its role as a pivotal technological force. In healthcare, AI is transforming the way diseases are diagnosed and treated. By leveraging advanced image analysis, AI improves the accuracy of disease detection [7]. Additionally, it aids in

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creating personalized treatment plans tailored to individual genetic profiles, revolutionizing patient care. In the financial sector, AI is employed for intricate tasks like risk assessment and fraud detection. By analyzing complex patterns, AI systems can predict market trends and identify potential financial risks, thereby enhancing the efficiency and security of financial operations.

AI's influence is also evident in the automotive and agricultural sectors. In automotive, AI is crucial in developing autonomous driving technologies, processing vast sensory data to enable real-time decision-making [8]. Meanwhile, in agriculture, AI is used for tasks ranging from drone-based crop monitoring to predictive analytics for yield optimization [9]. These applications underscore AI's adaptability in tackling various industry challenges. Additionally, AI's impact extends to environmental science and space exploration. In environmental science, AI facilitates climate modeling and helps predict weather patterns and climate change impacts more accurately [10]. In space exploration, AI processes astronomical data, aiding in the analysis of images and signals from space missions and contributing to discoveries about celestial bodies [11]. This wide-ranging application of AI across diverse fields showcases its transformative potential and capacity to address multifaceted challenges.

However, the rapid adoption of AI in various sectors has led to an increasing focus on its ethical dimensions, particularly concerning biases. While AI holds transformative potential, its reliance on existing data can result in the inadvertent replication of societal biases, an issue receiving significant attention globally.

In response to the rapid advancements and potential risks associated with AI, the European Union (EU) has established regulatory frameworks like the General Data Protection Regulation (GDPR) and the Digital Services Act to guide AI use [12]. To this end, the EU's AI Act categorizes AI systems based on risk, aiming to control potential harms, including biases [13].

In the United States, while there is no comprehensive federal law specifically for AI, significant steps have been taken toward responsible AI deployment. For instance, the White House Office of Science and Technology Policy (OSTP) released the Blueprint for an AI Bill of Rights in October 2022. This document provides guidance on designing, developing, and deploying AI to protect civil rights, focusing on safety, non-discrimination, privacy, transparency, and access to human alternatives in automated decision-making [14]. Additionally, in October 2023, President Biden issued an executive order to address potential societal harms from AI, including biases, emphasizing the need for strategies to identify and mitigate these issues [15].

Nonetheless, the challenge lies in the fact that AI creators and decision-makers often fail to address these biases adequately. The lack of transparency in AI algorithms and the absence of diverse perspectives in AI development teams further exacerbate the problem. Consequently, there is a growing call for more responsible AI development, which involves ethical considerations, transparency in AI systems, and the inclusion of diverse datasets and perspectives to mitigate inherent biases. This approach is crucial to harnessing AI's potential without inadvertently reinforcing societal inequalities [16].

Given these challenges, there is a growing consensus among society, researchers, and practitioners about the need to develop AI systems with ethical considerations at the forefront. This includes increasing diversity in datasets used for training AI models, employing techniques like algorithmic auditing to detect and correct biases at the development phase, and ensuring transparency in AI algorithms [17]. To fully address these ethical concerns, it is also important to include teams from different fields, such as ethicists, sociologists, and domain experts, in the development and decision-making processes of AI [18].

In this paper, we introduce the AI-Bias Network, a framework that thoroughly examines AI-enabled systems by highlighting the complex interplay of elements contributing to bias. To this end, we focus on the critical roles individuals play in the decision-making, design, and development stages. Our exploration is grounded in a comprehensive literature review and the integration of diverse theoretical frameworks, deliberately avoiding the oversimplification of bias within these intricate systems. Recognizing the profound impact of AI on organizations, societies, and individuals [19], we thoroughly examine the stages of AI model development to critically analyze potential sources of bias. This network visualization and analysis of AI-Bias Network shows the different types of bias that might affect AI's ability to make objective decisions, feedback, and results, which is what we want to happen: outcomes that are free of bias.

The organization of this paper is structured as follows: The methodology section details the systematic mapping process, outlining the research questions, search and study selection procedures, and data extraction and analysis methods. The results section presents the findings from the mapping study and addresses the research questions. This is followed by the discussion section, which interprets the retrieved literature and provides context to the results. Finally, the conclusion section synthesizes the overall findings, highlighting the implications and potential future directions for research.

2 Methodology

In this study, we employ a multifaceted approach to explore and map bias in AI-enabled systems, integrating a comprehensive literature review with diverse frameworks. We begin by conducting an extensive review of existing literature on AI bias, decision-making, and development processes, drawing from academic articles to establish a foundational understanding of the topic. Following this, we identify key elements that contribute to bias in AI systems, including data collection, algorithm design, and deployment practices.

Next, we utilize these insights to construct the AI-Bias Network, a framework designed to illustrate the complex interplay of bias-contributing factors. This network is developed through an iterative process, incorporating feedback from experts in AI ethics, data science, and social sciences.

To empirically validate our framework, we demonstrate using AI models across various applications. These use cases involve assessing the models for potential biases at different stages of their lifecycle- from data preprocessing to post-deployment.

By combining theoretical literature with the network science approach, our methodology provides a holistic view of bias in AI systems, aiming to uncover the underlying mechanisms and propose actionable strategies for achieving bias-free AI outcomes. This integrative method ensures that our findings are robust, comprehensive, and applicable across different AI contexts.

2.1 Literature review

This study adheres to the guidelines proposed by [20] for conducting systematic mapping reviews, which involve surveying a research area, classifying findings, and quantifying scholarly contributions within established categories. The primary objective of a systematic mapping review is to address broad research questions within a specific domain, uncover trends in existing research, and identify gaps in knowledge [20, 21]. This method is particularly advantageous when comprehensive overviews of the topic under investigation are lacking [22].

Systematic mapping studies follow a structured process encompassing the planning, execution, and reporting of findings. The planning phase involves formulating the research questions and devising a strategy for identifying relevant literature. During the execution phase, relevant papers are located through systematic searches, followed by a rigorous filtering and selection process based on titles and abstracts. This step, known as keywording, ensures that only pertinent studies are considered. Subsequently, the selected articles undergo thorough screening for data extraction. Finally, the

collected data is synthesized and mapped to provide a comprehensive overview of the research landscape.

2.2 Research questions

Following the guidelines for systematic mapping studies, the primary aim of this research is to identify potential bias activation points and construct a comprehensive, controllable framework to guide decision-makers and practitioners in the deployment of AI systems. This leads to the formulation of several research questions:

- **RQ1: In which parts of AI systems might biases emerge?** This question aims to pinpoint the specific stages in AI development and deployment where biases are likely to be introduced.
- **RQ2: What are the main sources and types of bias in AI systems as identified in the literature?** This question focuses on categorizing the various sources and types of biases, such as data bias, algorithmic bias, and user bias, documented in existing research.
- **RQ3: How do different theoretical frameworks and methodologies address bias in AI systems?** This question seeks to explore and compare the effectiveness of various theoretical approaches and methodologies in mitigating bias.
- **RQ4: What strategies and best practices are recommended for minimizing bias in AI systems?** This question aims to identify and evaluate strategies and best practices proposed in the literature to reduce bias and enhance fairness in AI applications. By seeking answers to these research questions, we aim to systematically map and classify the origins of bias in AI systems and highlight prominent categories and subcategories that are particularly susceptible to bias. In the following subsections, we detail the search protocol and methodological steps employed in this study.

2.3 Search

To address the research questions, two types of searches were conducted. First, a formal search was performed using established databases, namely ACM Digital Library, Web of Science, and Google Scholar. These databases were selected for their extensive collections of peer-reviewed work, including recent academic journals and conference papers relevant to AI and its associated biases. Web of Science, in particular, includes a significant number of AI-related publications, such as those from IEEE, providing a comprehensive source of literature. Advanced search options in both

Table 1 Search query samples

Queries
("AI" OR "Artificial Intelligence" OR "Machine Learning" OR "ML") AND ("Bias" OR "Biases" OR "Fairness" OR "Ethics")
("Artificial Intelligence" OR "AI" OR "Machine Learning" OR "ML") AND ("Ethical Implications" OR "Ethics" OR "Moral" OR "Fairness" OR "Bias")
("Artificial Intelligence" OR "AI" OR "Machine Learning" OR "ML") AND ("Algorithmic Bias" OR "Systemic Bias" OR "Bias in Algorithms" OR "Algorithmic Fairness")
("Artificial Intelligence" OR "AI" OR "Machine Learning" OR "ML") AND ("Bias Mitigation" OR "Bias Reduction" OR "Fairness" OR "Equity" OR "Bias Prevention")
("Artificial Intelligence" OR "AI" OR "Machine Learning" OR "ML") AND ("Bias in Development" OR "Bias in Deployment" OR "Developmental Bias" OR "Deployment Bias")
("Artificial Intelligence" OR "AI" OR "Machine Learning" OR "ML") AND ("Impact of Bias" OR "Bias Outcomes" OR "Consequences of Bias" OR "Bias Effects")

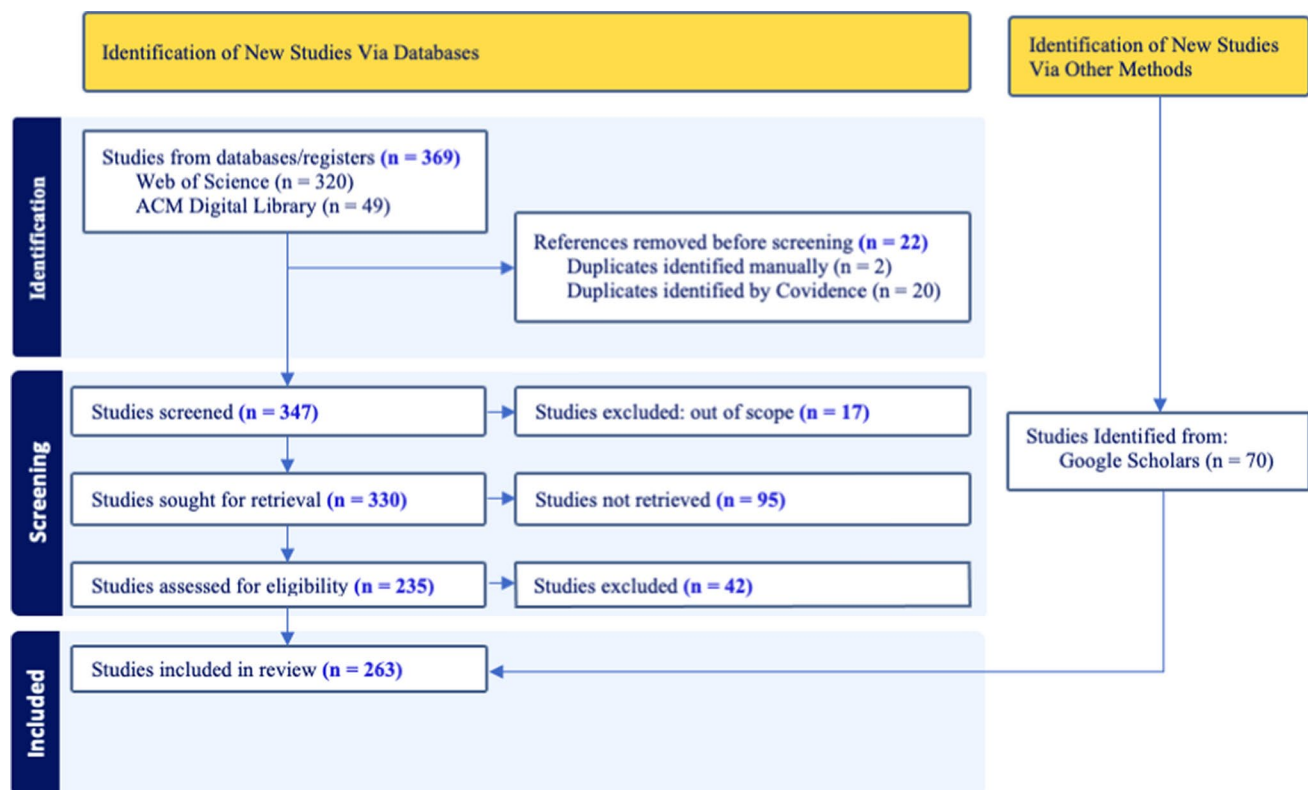
databases allowed for the use of keywords and the application of filters.

The search was limited to articles written in English and published within the last five years (2019–2023). This time frame was chosen to ensure that the review included the most current and relevant scholarly work, reflecting the rapid advancements and increasing attention to bias in AI systems over recent years.

To ensure inclusivity and avoid bias in the search results, a combination of general keywords and specific search criteria was employed. Keywords used included terms like "bias," "biases," "artificial intelligence," "AI," "machine learning," and "ML," with Boolean operators to incorporate synonyms and refine results. These keywords are listed in Table 1 as samples; additional related terms were also used in the searches.

The rationale for focusing on the past five years lies in the significant advancements and increased scrutiny regarding bias in AI during this period. As AI technologies have rapidly evolved, the recognition and understanding of their potential biases have also grown, making recent literature particularly relevant to this study. Although a five-year timeframe might seem short, the broad categories of bias discussion-encompassing data, development, and human-AI teaming/interaction-effectively capture the current landscape. Our work incorporates these discussions while also acknowledging the historical context of AI bias, which dates back to the 1980 s [23]. This approach ensures that our study covers the most critical and contemporary aspects of AI bias while grounding it in its historical evolution.

Following the formal search process, we utilized the advanced search features in Google Scholar to refine the understanding of *bias* terms found in the articles sourced from ACM and Web of Science. Figure 1 illustrates all

**Fig. 1** Prisma flow diagram displaying the steps taken during systematic literature review

articles retrieved from these databases, including Google Scholar.

Please note that, in this project, we utilized Covidence [24], a reference management tool designed to streamline the systematic review process, to efficiently manage and screen the large volume of articles retrieved from multiple databases. The tool automatically identified and removed duplicate entries, ensuring a clean dataset for analysis. Covidence also facilitated the manual screening process, helping us to assess the eligibility of each paper based on predefined inclusion and exclusion criteria, thereby enhancing the overall accuracy and efficiency of our review process.

During the initial screening of the 369 papers retrieved, 20 duplicate papers were automatically identified and removed with 2 additional duplicates eliminated manually. The titles and abstracts of the remaining 347 papers were then screened for eligibility, focusing on their evaluation of bias in AI systems. Papers not meeting these criteria, as detailed in Table 2, were excluded, resulting in 17 excluded papers.

We aimed to retrieve the full text of all relevant papers during the extraction process. We successfully obtained the full text for 235 out of 330 papers. After this, we dropped 42 additional papers as these were irrelevant to our research. Finally, we conducted an informal search to enhance our understanding of AI-related bias concepts, which allowed us to include 70 publicly accessible publications. This approach ensured that our review encompassed a comprehensive range of studies, incorporating both formally indexed and openly available resources, adding to the total of 263 papers in our review process.

In our categorization of bias into Data, Design, and Human-AI Interaction, we followed a grounded approach based on the literature review. The process began by screening the papers to identify how each study framed and discussed bias. Many papers explicitly focused on one area of bias, such as bias in data or design, while others examined how bias emerged within specific contexts, such as during model training or deployment. By identifying

commonalities in these discussions, we were able to group the biases into three primary categories. This categorization should be viewed as an initial framework, capturing the most frequently discussed sources of bias across the studies we reviewed. However, we acknowledge that bias in AI systems is complex and may arise in additional contexts, such as in the application of AI in specific domains (e.g., border control or policing). While our framework does not claim to be exhaustive, it provides a robust starting point for understanding bias activation areas. Additional areas of bias can be incorporated into this framework as they emerge from future studies or specific applications, making it adaptable and extendable as the field evolves.

The collected papers and data were organized into a table (see Appendix A), detailing the bias activation points identified in AI systems, the domain of publication, and the type of article. Through a full-text review, we noted references to biases and their emergence across various domains.

From a systemic perspective, we identified three primary origins of bias activation: Data, Design, and human-AI teaming (HAI) (also called human AI interaction). In the following section, we demonstrate the dynamic entanglement of these activation points, indicating that they cannot be studied in isolation. Subsequently, we created a network graph to visualize activation paths according to AI system domains. This comprehensive approach highlights that addressing bias requires the cooperation of all decision-makers and key players, acknowledging that some important activation points within the system might have been overlooked.

3 Findings from systematic review

In this section, before proceeding to the AI-Bias network, we would like to highlight some of the findings we have observed in conducting the systematic literature review.

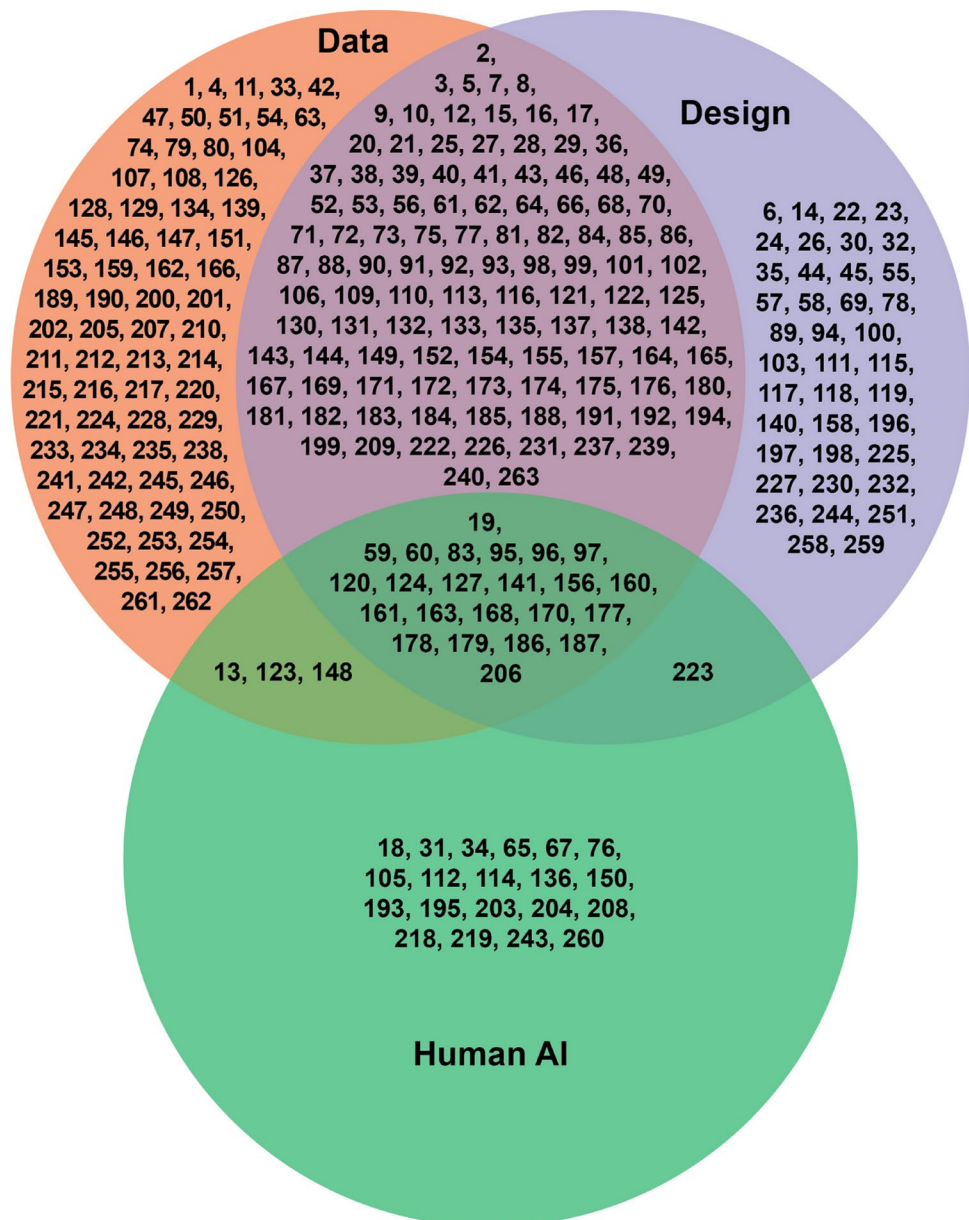
After screening the full text of 263 papers, we identified several prevalent themes and specific categories where bias can arise within AI systems. Notably, 77.18% of the articles addressed biases that can emerge from data, while 65.02% focused on biases originating in the design process. In contrast, only 17.87% of the articles discussed bias activation in human-AI interaction. These findings highlight the significant attention given to data and design biases in the existing literature, underscoring the critical need for comprehensive strategies in these areas.

Figure 2 illustrates the relationships between the sets of articles and their overlapping coverage of bias activation points. The Venn diagram shows significant intersections, indicating that many papers discuss biases across multiple categories: Data/Design, Data/HAI, Design/HAI, and the

Table 2 Screening publications for In/exclusion criteria

Criteria	Details
Inclusion criteria	Studies addressing bias in AI data, models, or human-AI interaction Research covering bias across various domains Peer-reviewed articles, conference papers, and technical reports
Exclusion criteria	Irrelevant topics Focus on AI as a tool to mitigate bias rather than sources of bias Papers that endorse or praise bias Human bias against AI Purely statistical studies without practical implications

Fig. 2 Articles discussing bias, Numbers refer to the table in Appendix A



intersection of all three. Articles covering bias in both Data and Design demonstrate the highest overlap, reflecting the intertwined nature of these two aspects. Conversely, human-AI teaming and interaction are the least covered, revealing a gap in the literature despite their importance in addressing bias impacts. This gap suggests a need for further research and targeted efforts to address biases in human-AI interaction comprehensively.

Journal articles tackled bias issues in AI systems with over 75% higher rate compared to conference articles. Another observation revealed that some studies mentioned the same type of bias activation points but in different contexts and across various activation parts of AI systems. For instance, confirmation bias impacted the data collection process in one case where the collected data conformed to the

perspective of the data collector [25]. In another, confirmation bias adversely affected the human-AI teaming [26].

4 AI-bias network

In this section, we introduce a framework for parameterizing each bias-inducing factor within the AI-enabled ecosystem. Following our systematic mapping review to identify research trends and establish our area of study in exploring bias manifestation within AI systems [27], we draw inspiration from the rhizomatic framework proposed by Deleuze and Guattari [28]. Bias in AI does not conform to a hierarchical structure; instead, it mirrors the rhizome's non-linear characteristics, connecting disparate points in an unordered,

non-hierarchical fashion. This analogy effectively illustrates the complex and chaotic nature of bias propagation within AI models [29].

In our approach, the rhizome concept emphasizes how bias in AI can activate, amplify, and evolve through various dimensions of the system, similar to interconnected roots. Bias possesses determinations (connections activating biased results), magnitudes (degree of activation), and dimensions (number of connections across layers), reflecting the rhizome's principles of multiplicity and interconnectedness. Bias can emerge from any point and propagate across different layers, and eliminating bias in one element doesn't guarantee its removal due to the rhizome's principle of rupture—it may reappear through other connections. The experimental nature of bias mapping underscores that bias is dynamic and challenging to trace within AI's “black box.” Thus, the rhizomatic perspective helps us understand bias as a complex, evolving phenomenon intersecting with the network-based approach by illustrating how it spreads across interconnected nodes without a clear hierarchical structure [28, 30].

Figure 3 illustrates the network of bias sources in AI-enabled systems, depicting the complex interconnections among various factors contributing to bias in data, design, and human-AI teaming. Our AI-Bias Network seeks to encapsulate this complexity by mapping the diverse sources and pathways through which bias can emerge and propagate in AI systems. By recognizing the interconnected and dynamic nature of these bias sources, we aim to provide a

comprehensive tool for understanding and mitigating bias in AI. This approach underscores the need for holistic strategies that consider the multifaceted and intertwined origins of bias, promoting more equitable and transparent AI practices.

The AI-enabled systems, structured and ordered, diverge fundamentally from a rhizome. However, the biases within such a system embody rhizomatic attributes: multiplicity, where various factors and relationships contribute to biased outcomes; magnitude, indicating the extent or degree of bias activation; and dimensionality, denoting the network of interconnections among the system's layers or multiplicities [31]. These biases create a “zone of indiscernibility,” where boundaries between multiplicities are not clearly defined, complicating the demarcation and isolation of biased components within the AI's structure [30].

Decision-making processes within AI systems act as regulatory forces over these rhizomatic biases, determining the scope and scale of their influence. Furthermore, the rhizome's principle of rupture is evident, as addressing a single element does not eradicate biases; they can re-emerge through other connections within the system, whether existing or newly formed [29].

Thus, bias within AI systems lacks fixed structure and stability, rendering it elusive within the algorithmic “black box.” Nevertheless, it can be mapped experimentally, drawing from empirical reality, as evidenced by the connections and attributes identified in the provided network diagram [28, 30]. Figure 3 presents a comprehensive landscape

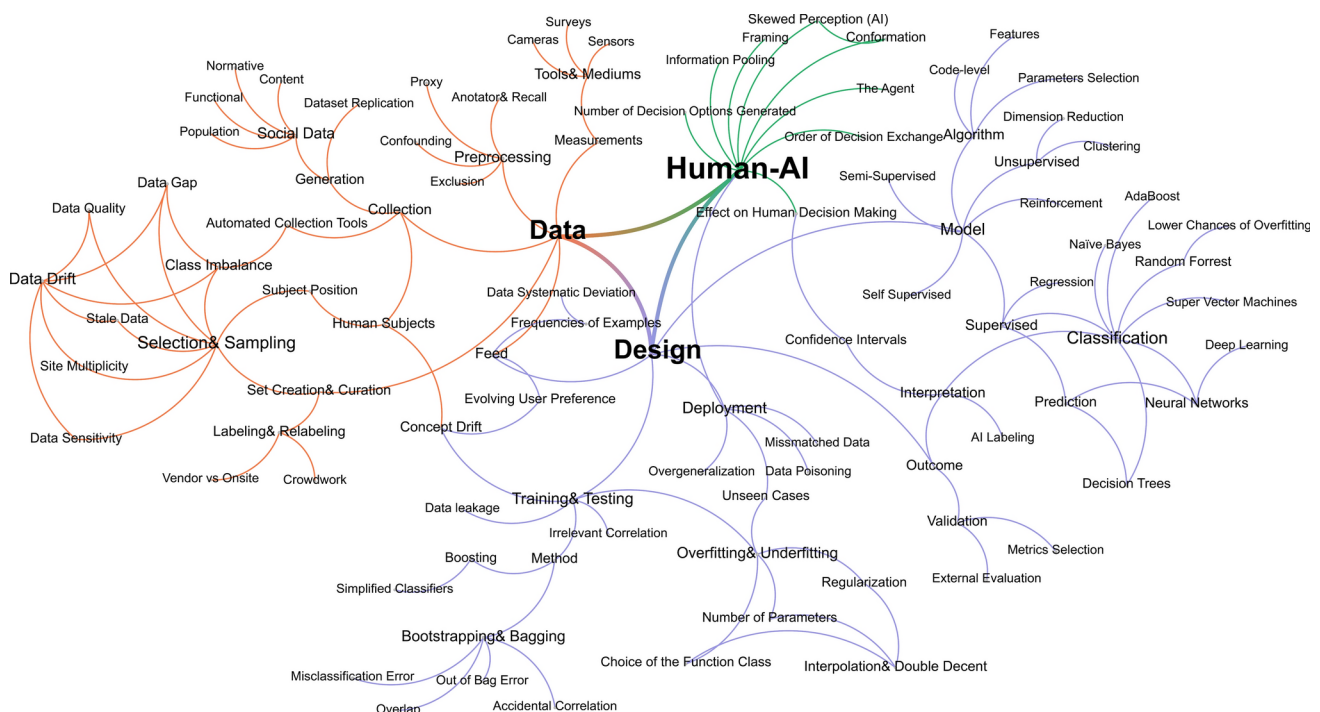


Fig. 3 The network of sources of bias in AI-enabled systems [Please see Appendix for Larger version.]

encapsulating the various elements encompassing Data, Design, and Human-AI teaming dependencies.

This initiative culminated in the development of a structured network that maps a diverse array of elements potentially contributing to bias in AI outcomes. Central to this network are three key components known to influence AI bias: Data, Design, and human-AI teaming. Data encompasses biases present in AI training datasets, including historical biases [32], sampling biases [33], and data gaps [34]. Design addresses the construction of AI algorithms, focusing on aspects such as feature selection [35], model complexity [36], and optimization methods [37]. Human-AI Teaming highlights the impact of biases from various stakeholders, including developers [38], annotators [32], and end-users [7].

This comprehensive network framework provides a nuanced understanding of how these components interact and contribute to bias in AI systems. By identifying and mapping these bias-inducing factors, the framework offers valuable insights into effective strategies for mitigating biases, ultimately promoting fairness and transparency in AI outcomes. This approach underscores the necessity of a holistic perspective that considers the multifaceted and interconnected nature of bias in AI, ensuring more equitable and reliable AI practices.

In practical terms, ownership of each bias activation area may vary depending on the context and stakeholders involved. Bias in Data may be best owned by data scientists and domain experts responsible for curating, collecting, and ensuring the diversity and representativeness of the dataset. For bias in Design, ownership could lie with AI developers and engineers, who must ensure the fairness and transparency of algorithms. Bias in Human-AI Interaction may fall under the purview of end-users, human-computer interaction specialists, and ethicists who assess the impact of AI systems on human behavior and decisions.

The division of ownership across these areas has important implications for bias mitigation. If ownership is fragmented, coordination among the different stakeholders is crucial to avoid a siloed approach where biases in one area are addressed but inadvertently reinforced in another. A collaborative approach involving cross-functional teams that include data scientists, developers, ethicists, and domain experts is therefore essential to ensure comprehensive bias mitigation throughout the lifecycle of AI systems. This ownership distribution encourages accountability, where each stakeholder is responsible for addressing bias within their specific domain while remaining aligned with the overall goal of creating fair and equitable AI systems.

5 Network science perspective

The network analysis reveals that the human-AI teaming (HAI) node exhibits the highest degree of connectivity, indicating that biases activated at this stage are intricately linked to various aspects of the system. The edge colors in the network denote different dependencies: connections to Data are shown in orange, to Design in purple, and to human-AI teaming in green. Despite the fewer green lines representing human-AI teaming, the high degree of this node (11) signifies its central role and the complex interplay between Data and Design features with human-AI teaming.

The high degree of the HAI node is due to its direct involvement in human interactions, where biases often emerge from system-user interactions. This is contrasted by the rest of the network, which displays a more tree-like structure, emphasizing the direct and significant influence of HAI on bias.

The network's high modularity score (0.742) reflects a dense clustering of connections within categories and sparse associations between features across different modules. This indicates that biases are strongly interconnected within specific areas, but less so between different bias categories. The presence of a high number of communities (12), given the total number of nodes (100) and dependencies (3), underscores the complex structure of connections in biased AI systems. This complexity highlights the need for comprehensive approaches to understand and mitigate bias, considering the multifaceted interactions within the network.

6 Software engineering perspectives

6.1 Data dependency

Data is the lifeblood of AI systems, fundamentally shaping their behavior and outputs, as displayed in the “Data” node and its sub-branches in Fig. 3. The processes of collecting, refining, and selecting data are critical in shaping the impact of AI technologies [32, 34]. The quality [39], diversity, and representativeness of the data determine the AI's performance reliability [40]. High-quality data is similar to using superior bricks in construction, providing a strong foundation for model building. The diversity of data encompasses a broad spectrum of scenarios, mirroring the multifaceted nature of the real world, while representativeness ensures that the AI system's outputs accurately mirror the intended domain, thus avoiding skewed results and inherent biases.

The introduction of biases at various stages of the data pipeline warrants attention [41]. Sampling methods [33, 34, 42–44], human judgment in data collection, and the inherent characteristics of the data can lead to distortions in the AI's

learning and decision-making processes. An overemphasis on specific demographics or patterns can inadvertently lead to the reinforcement of societal inequalities [45]. To mitigate these biases, meticulous data preprocessing and curation are imperative. Techniques such as data cleaning, balancing, and augmentation not only improve data quality but also ensure its representativeness. Feature engineering is employed to refine data, optimizing it for training efficient AI models. Nonetheless, subsetting data for particular tasks requires wise consideration to avoid introducing new biases and to preserve the representativeness of the larger dataset.

6.2 Design dependency

The “Design” node in Fig. 3 shows an important part of AI systems: the architecture of algorithms. This is what controls how decisions are made and how outcomes might turn out [46]. These algorithmic frameworks, which are like architectural blueprints, affect more than just computation [47]. They can also change the way society works and introduce biases that can make existing problems worse or create new ones. Thus, the challenge in AI design transcends technical adeptness, demanding a commitment to ethical integrity [37].

Algorithmic propensities to bias are not uniform; certain models, such as linear regression, might unwittingly escalate pre-existing biases present in the data they process, potentially leading to discriminatory results. In contrast, models like decision trees may exhibit an enhanced tolerance against such biases [37]. Recognizing the differential bias susceptibility of algorithms is vital for developers and designers, steering them toward the adoption of models that are ethically aligned as well as functionally adept [48].

Additionally, the design phase stands out as a pivotal stage for the strategic mitigation of bias [47]. The selection of transparent algorithms, such as interpretable models, enables scrutiny and potential debiasing of AI logic. In addition to research efforts aimed at the development of intrinsically fair algorithms, techniques like data augmentation and counterfactual analysis are essential for cleaning training datasets [49].

Yet, the quest to neutralize bias must maintain AI efficacy. A delicate balance is required, achieved through rigorous testing and continuous algorithmic refinement, with a collaborative approach among developers, designers, researchers, and policymakers. Open discussion and the sharing of best practices serve as the foundation for this collective action, facilitating ongoing algorithmic assessment and improvement [48].

Ultimately, by thoroughly engaging with the nuances of algorithmic design, we can pave the way for AI that is not only powerful but also principled. This will lead to the

creation of inclusive technologies that advance the collective good and set a benchmark for future technological innovations.

6.3 User (human) dependency

The “Human-AI Teaming” node in Fig. 3 reveals the subtle yet significant impact of human influence on AI development. The biases of designers, developers, and users can subtly weave into AI-enabled systems, affecting their output and societal impact. Biases can originate from the preconceptions of designers, the technical choices of developers, or the interaction patterns of users [7]. The interpretation of AI recommendations, such as news feeds, can also perpetuate biases, reinforcing user echo chambers.

To counteract this, we must acknowledge the human factor in AI creation. Diversity in development teams, thorough testing for biases, and transparency in AI processes are vital [18]. We need to view humans not just as end-users but as integral contributors to AI’s evolution. Building responsible AI is an ongoing process that requires a change in our interaction with technology [50]. By recognizing and addressing the human influence on AI, we aim for a future where AI upholds the best of human values, fostering an equitable and just society [51].

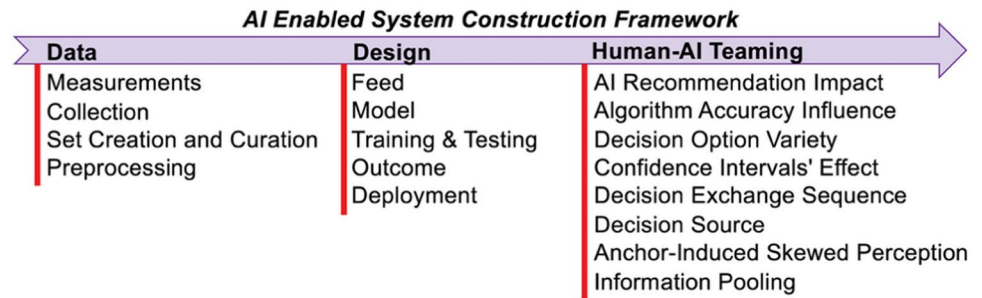
6.4 Societal and cultural dependencies

AI systems are deeply intertwined with the broader social and cultural fabric, often reflecting prevailing biases and stereotypes [52]. While these influences may not be overtly depicted in our diagrammatic representation, their implicit presence is evident in elements such as data collection processes and the dynamics of human-AI interaction.

The entrenchment of societal norms within AI [41, 53] underscores the necessity for an integrated development strategy. This strategy should include incorporating a broad spectrum of data to enhance the diversity of inputs informing AI behavior. Furthermore, it is crucial to design algorithms that prioritize ethical soundness alongside technical efficacy. Ethical considerations must be integrated throughout the AI development lifecycle and supported by continuous bias monitoring. By adhering to these practices, AI systems can be guided towards greater impartiality and fairness, thereby contributing to the creation of more equitable technological ecosystems.

In Fig. 4, we outline essential steps aimed at reducing bias in the design and implementation phases of AI-enabled systems. This foundational guideline is accessible to both technical and non-technical individuals, facilitating the development of more equitable AI solutions.

Fig. 4 The theoretical framework of Bias landscape for deploying AI-enabled systems



Within the Data plane, ensuring diversity and employing robust sampling techniques are crucial for obtaining representative datasets. This is complemented by thorough data cleaning. Set creation and curation involve blind labeling and debiasing techniques, while preprocessing focuses on feature engineering and normalization to prevent skewness [32, 54]. During the usage phase, continuous monitoring, transparency, and human oversight are vital for addressing biases in AI outputs. Each stage, from data handling to deployment, requires vigilance and adaptability to foster fair and equitable AI systems [55].

In the Design phase, it is pivotal to use diverse data sources, careful feature engineering, selective algorithm choice, and fairness-aware evaluation metrics [16, 36, 56]. The Feed phase emphasizes data cleaning, balancing, and augmentation for unbiased training. In the Model phase, the focus shifts to fairness in training and bias testing. The Training & Testing phase involves refining the model through feedback loops to enhance fairness [37]. Finally, the Outcome phase stresses continuous monitoring, evaluation, and transparency to identify and rectify biases, underscoring the iterative nature of creating equitable AI systems.

Understanding factors such as AI recommendation impact, algorithm accuracy, decision option variety, confidence intervals, decision exchange sequence, decision source, anchor-induced skewed perception, and information pooling is crucial for mitigating bias in AI-assisted decision-making [7, 48, 57–59]. High reliance on AI recommendations or algorithmic accuracy can amplify biases, while critical engagement, diverse options, and transparent decision-making processes can counteract bias. Recognizing these dynamics allows human-AI teams to foster critical analysis and diverse perspectives, promoting fairer outcomes in AI systems.

In summary, this systematic framework represents a promising approach to addressing bias in AI-enabled systems. Further research and development are necessary to validate its practical effectiveness and refine its implementation for specific tasks and datasets. Integrating this framework with human expertise and ethical considerations could significantly contribute to the development of fairer and more responsible AI.

7 AI-bias network use cases

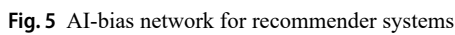
In this section, we explore practical applications of the AI-Bias Network framework across various domains. By examining specific use cases, we illustrate how the framework can identify, analyze, and mitigate biases in AI systems. These examples demonstrate the versatility and applicability of the AI-Bias Network in real-world scenarios, providing insights into its effectiveness in promoting fairness and equity in AI implementations. Through detailed analysis, we aim to highlight the critical role of the AI-Bias Network in addressing biases that arise in different contexts, from healthcare to recommendation systems, thereby underscoring its importance as a tool for developing more ethical and responsible AI technologies.

7.1 Recommendation systems

Recommendation systems are AI-driven tools designed to suggest items to users based on their preferences, behaviors, and interactions. These systems are widely used in various domains, including e-commerce, streaming services, and social media, to personalize user experiences by recommending products, movies, music, or content that aligns with their interests. Figure 5 illustrates the principal bias-causing reasons effective in Recommendation Systems.

In the context of Recommendation systems, several types of biases can be introduced at different stages of data collection and algorithm design. Inductive biases may be deliberately incorporated by researchers to improve model performance. However, biases can also emerge through user interactions with the recommendations. For instance, selection and exposure bias arise from imbalanced observations, where certain items receive more attention than others, leading to skewed data. Conformity bias occurs when users' preferences deviate due to a tendency to engage in similar behaviors as others, while position bias is evident when users interact more frequently with items listed at the top of recommendations.

Human-AI interactions contribute significantly to these biases, as user behaviors and preferences continuously feed into the model, creating new entries that reflect these biases.



Understanding these biases and their sources is crucial for developing fairer and more equitable recommendation systems. It requires a holistic approach that addresses biases at every stage, from data collection to algorithm design and user interaction.

The literature indicates that the use of AI in healthcare is a significant concern for practitioners and decision-makers, as it can impact their judgment in diagnosis and prescribing medications due to cognitive and automation biases introduced by AI systems [61, 62]. Figure 6 illustrates the main bias-causing reasons effective in healthcare personalized medicine projects. Training models on unbalanced datasets that overlook marginalized communities, are selective, misrepresentative, or contain missing data reinforces biases within the system [63]. Algorithmic design choices, such as employing unsupervised or self-supervised learning methods, ignoring confounding factors, and generating selectivity without relevance, can activate biases within the design process [64]. Additionally, healthcare professionals

Biases in AI systems can result in serious consequences, particularly for protected groups, by mistakenly classifying them as healthy when they require care, thereby depriving them of timely medical attention. The healthcare sector, more than any other, necessitates close collaboration with AI developers to address these concerns and develop just AI systems. Healthcare practitioners and decision-makers are aware of the flaws in AI systems, such as issues with the data used to train these systems. They require the expertise of statisticians and computer scientists to adjust and fine-tune these systems for better outcomes.

AI bias has been a pervasive issue since the early days of machine learning, and while our paper may be just a drop in the vast ocean of AI bias research, it tackles a critical need: a systematic framework that identifies where, when, how, and why bias arises in AI systems. To develop this framework, we began by exploring the state-of-the-art in AI bias research, thoroughly reviewing 347 papers to understand the current landscape. This extensive review informed our approach and highlighted the gaps that our framework seeks to address.

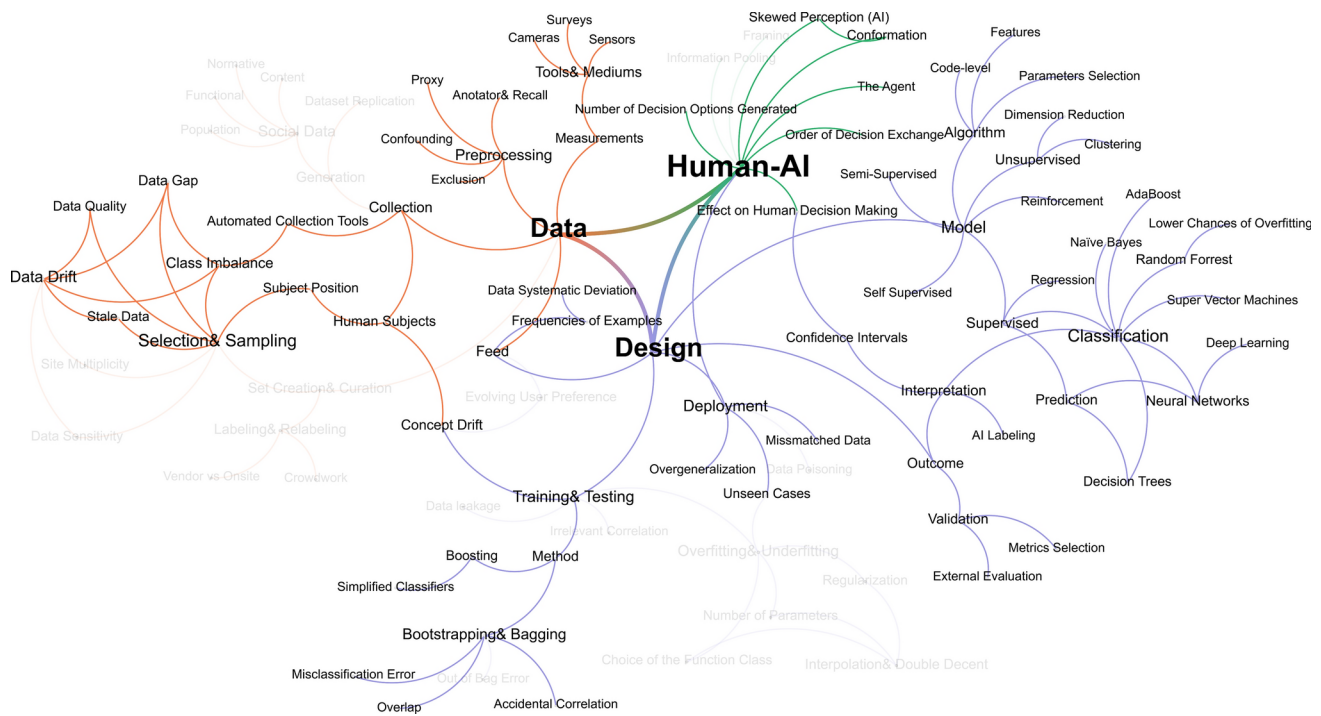


Fig. 6 AI-bias network for healthcare personalized medicine

Our proposed system design framework acknowledges that the various components of an AI model can respond differently and non-linearly depending on the specific objectives of the domains in which they are utilized. This nuanced understanding underscores the importance of informed decision-making and regulatory measures throughout the deployment of AI models, which are essential to effectively mitigating bias. This framework seeks to ensure that AI systems are not only technically proficient but also ethically responsible and socially considerate, balancing technical excellence with ethical integrity.

Moreover, in this paper, we have presented a comprehensive overview of the multifaceted aspects of AI-enabled systems, focusing on the complex interplay of bias-contributing elements and the involvement of individuals at various stages of AI development. Our study highlights the multiple layers of bias sources in AI, and we have developed an intricate network that maps out the diverse elements leading to bias, focusing on Data, Design, and Human-AI Teaming.

Additionally, by applying Deleuze and Guattari's concept of rhizome to mapping biases in AI, we proposed that the progression of bias in an AI-enabled system might be nonlinear. To examine this hypothesis, we recommend

designing a series of experiments to measure the relationship between variations in an AI model's error rate as a function of bias or variations in bias. The core idea is to measure the accuracy of an AI-enabled system before and after introducing bias in specific elements—Data, Design, or human-AI teaming. By activating bias in only one of these elements during each experiment, we aim to trace its impact and develop targeted mitigation strategies.

Looking forward, future research will involve incorporating a control model and real-world case studies to validate our framework and refine our approach based on empirical findings. Collaborating with various organizations will provide practical settings to test and improve our model. Furthermore, we plan to delve deeper into the ethical implications of AI bias mitigation and develop comprehensive guidelines for responsible AI development.

As AI continues to evolve, particularly with the rise of Generative AI and Foundational Models, our research aims to contribute significantly to understanding the societal impacts of AI systems. This work is a step toward advancing truly unbiased AI technologies, ensuring that the ongoing evolution in AI promotes fairness, transparency, and ethical responsibility.

Appendix A: The complete list of the reviewed literature

#	Author References	Type	In Domain	Bias_ Category
1	Abhari et al., 2023	conferencePaper	Healthcare	Data
2	Abrmoff et al., 2023	journalArticle	Healthcare	Data, Design
3	Abusitta et al., 2020	conferencePaper	Computer Science	Data, Design
4	Afrose et al., 2022	journalArticle	Healthcare	Data
5	Ahmed et al., 2021	conferencePaper	Computer Science	Data, Design
6	Akter et al., 2021	journalArticle	Business	Design
7	Akter et al., 2022	journalArticle	Business	Data, Design
8	Akter et al., 2021	journalArticle	Information Management	Data, Design
9	Akter et al., 2023	journalArticle	Marketing	Data, Design
10	Al-Luhaybi et al., 2022	conferencePaper	Computer Science	Data, Design
11	Alelyani, 2021	journalArticle	Computer Science	Data
12	Allareddy et al., 2023	journalArticle	Healthcare	Data, Design
13	Alon-Barkat et al., 2023	journalArticle	Public Administration	Data, HAI
14	Alves et al., 2021	conferencePaper	Computer Science	Design
15	Aquino et al., 2023	journalArticle	Healthcare	Data, Design
16	Aquino et al., 2023	journalArticle	Healthcare	Data, Design
17	Atay et al., 2021	conferencePaper	Computer Science	Data, Design
18	Bach et al., 2023	conferencePaper	Healthcare	HAI
19	Balayn et al., 2021	journalArticle	Information Systems	Data, Design, HAI
20	Barbosa et al., 2019	conferencePaper	Computer Science	Data, Design
21	Beattie et al., 2022	conferencePaper	Computer Science	Data, Design
22	Bekerman et al., 2022	conferencePaper	Math and Statistics	Design
23	Belitz et al., 2021	journalArticle	Computer Science	Design
24	Belkin et al., 2019	journalArticle	Computer Science	Design
25	Bellamy et al., 2019	journalArticle	Computer Science	Data, Design

#	Author References	Type	In Domain	Bias_ Category
26	Belzak, 2023	journalArticle	Educational Measurement	Design
27	Bernagozzi et al., 2021	conferencePaper	Computer Science	Data, Design
28	Besse et al., 2022	journalArticle	Math and Statistics	Data, Design
29	Blanzeisky et al., 2021	conferencePaper	Computer Science	Data, Design
30	Blanzeisky et al., 2022	journalArticle	Computer Science	Design
31	Boonprakong et al., 2023	conferencePaper	Computer Science	HAI
32	Booth et al., 2021	conferencePaper	Computer Science	Design
33	Braun et al., 2023	conferencePaper	BioInformatics	Data
34	Brown et al., 2023	journalArticle	Healthcare	HAI
35	Burlina et al., 2021	journalArticle	Healthcare	Design
36	Buslun et al., 2023	journalArticle	healthcare	Data, Design
37	Cabrera et al., 2019	conferencePaper	Computer Science	Data, Design
38	Calders et al., 2021	journalArticle	General	Data, Design
39	Cao et al., 2021	journalArticle	Computational Linguistics	Data, Design
40	Cardenas et al., 2019	conferencePaper	Social Science	Data, Design
41	Cha et al., 2021	journalArticle	Medical Informatics	Data, Design
42	Chen et al., 2023	conferencePaper	Communication	Data
43	Chen et al., 2023	journalArticle	Medical Informatics	Data, Design
44	Chen et al., 2023	journalArticle	Computer Science	Design
45	Chen et al., 2023	journalArticle	Medical Informatics	Design
46	Cho, 2021	journalArticle	Healthcare	Data, Design
47	Choudhury et al., 2020	journalArticle	Business	Data
48	Chu et al., 2023	journalArticle	Social Science	Data, Design
49	Chung et al., 2021	journalArticle	Psychology	Data, Design
50	Cirillo et al., 2020	journalArticle	Healthcare	Data

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#	Author References	Type	In Domain	Bias_Cat-egory
107	Kafkalias et al., 2022	conference-Paper	Information Processing	Data
108	Kanakala et al., 2023	journalArticle	Chemistry	Data
109	Katare et al., 2022	conference-Paper	Computer Science	Data, Design
110	Kim et al., 2023	journalArticle	Healthcare	Data, Design
111	Kiyasseh et al., 2023	journalArticle	Healthcare	Design
112	Kliegr et al., 2021	journalArticle	Psychology	HAI
113	Kronman, 2023	journalArticle	Humanities	Data, Design
114	Kupfer et al., 2023	journalArticle	Psychology	HAI
115	Lake et al., 2022	journalArticle	Math and Statistics	Design
116	Landers et al., 2023	journalArticle	Psychology	Data, Design
117	Langer et al., 2023	journalArticle	Business	Design
118	Lewicki et al., 2023	conference-Paper	Computer Science	Design
119	Liang et al., 2020	conference-Paper	Information Studies	Design
120	Lobo et al., 2023	journalArticle	Knowledge Media	Data, Design, HAI
121	Lwowski et al., 2021	journalArticle	Medical Informatics	Data, Design
122	Marinucci et al., 2023	journalArticle	Psychology	Data, Design
123	McCradden et al., 2020	journalArticle	Healthcare	Data, HAI
124	Mehrabi et al., 2021	journalArticle	Computer Science	Data, Design, HAI
125	Melsiun et al., 2021	conference-Paper	Computer Science	Data, Design
126	Miceli et al., 2022	journalArticle	Computer Science	Data
127	Mittermaier et al., 2023	journalArticle	Medical Informatics	Data, Design, HAI
128	Momenzadeh et al., 2022	journalArticle	BioInformatics	Data
129	Monroe-White et al., 2023	journalArticle	Business	Data
130	Mosteiro et al., 2022	journalArticle	Information Science	Data, Design
131	Muralidhar, 2021	conference-Paper	Computer Science	Data, Design
132	Nadeem et al., 2022	journalArticle	Information Systems	Data, Design
133	Narayanan Venkit et al., 2023	conference-Paper	Computer Science	Data, Design
134	Navarro et al., 2021	journalArticle	Healthcare	Data
135	Norori et al., 2021	journalArticle	Healthcare	Data, Design

#	Author References	Type	In Domain	Bias_Cat-egory
136	Nourani et al., 2021	conference-Paper	Computer Science	HAI
137	O'Connor et al., 2023	journalArticle	Politics	Data, Model
138	Oduoye et al., 2023	journalArticle	Healthcare	Data, Design
139	Openja et al., 2024	journalArticle	Computer Science	Data
140	Pagano et al., 2023	journalArticle	Computer Science	Design
141	Pagano et al., 2023	journalArticle	Computer Science	Data, Design, HAI
142	Pandey et al., 2021	conference-Paper	Computer Science	Data, Design
143	Pastaltzidis et al., 2022	conference-Paper	Computer Science	Data, Design
144	Paul et al., 2022	journalArticle	Healthcare	Data, Design
145	Pavun Perez et al., 2023	conference-Paper	Finance	Data
146	Peoa et al., 2020	conference-Paper	Computer Science	Data
147	Penedo et al., 2023	journalArticle	Finance	Data
148	Pot et al., 2021	journalArticle	Healthcare	Data, HAI
149	Rana et al., 2023	journalArticle	Computer Science	Data, Design
150	Rastogi et al., 2022	journalArticle	Computer Science	HAI

#	Author References	Type	In Domain	Bias_Cat-egory
151	Rouzrokh et al., 2022	journalArticle	Healthcare	Data
152	Rueckel et al., 2021	journalArticle	Healthcare	Data, Design
153	Seyyed-Kalantari et al., 2021	journalArticle	Healthcare	Data
154	Shrestha et al., 2022	journalArticle	Computer Science	Data, Design
155	Silva et al., 2022	conferencePaper	Law	Data, Design
156	Soboczenski et al., 2019	journalArticle	Medical Informatics	Data, Design, HAI
157	Soleimani et al., 2022	journalArticle	Business	Data, Design
158	Solyst et al., 2023	journalArticle	Computer Science	Design
159	Spisak, 2022	journalArticle	Statistical Arts	Data
160	Srinivasan et al., 2021	conferencePaper	Arts	Data, Design, HAI
161	Srinivasan et al., 2021	journalArticle	Information Science	Data, Design, HAI
162	Starke et al., 2021	journalArticle	Philosophy of Science	Data
163	Strauss, 2021	journalArticle	Information/Social Science	Data, Design, HAI

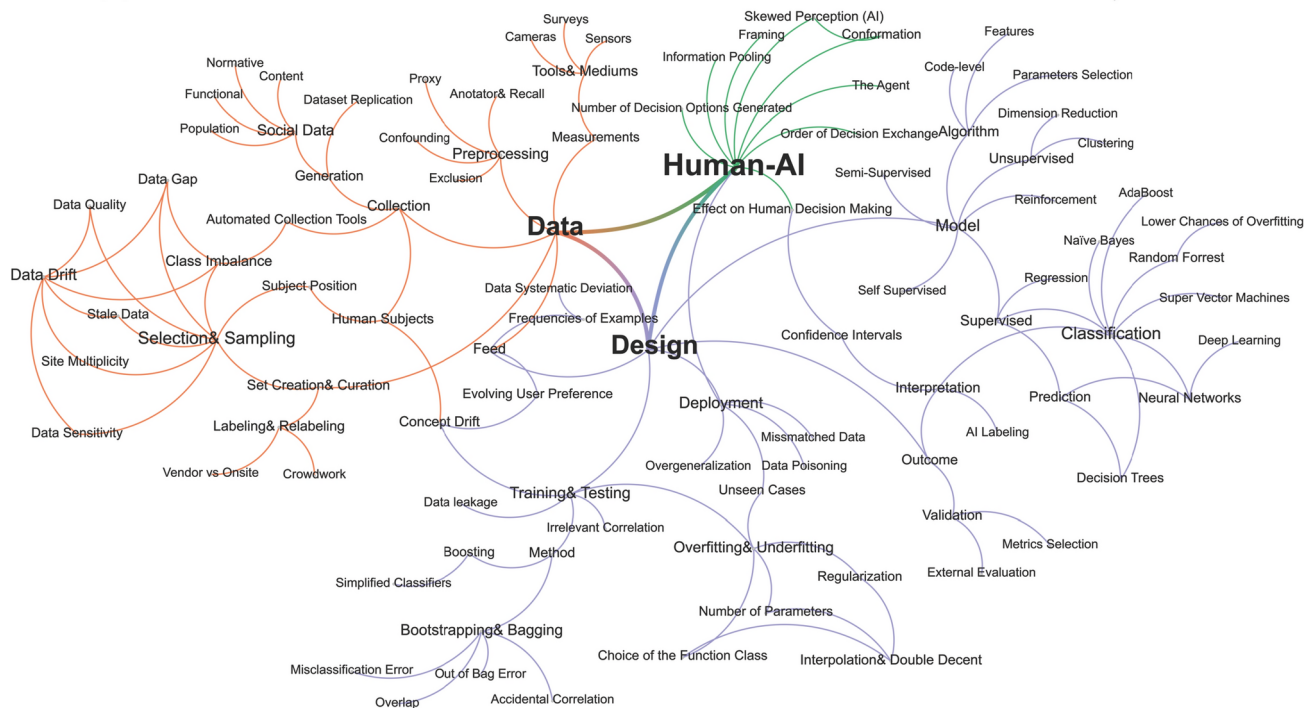
#	Author References	Type	In Domain	Bias_Cat- egory
164	Straw et al., 2020	journalArticle	Healthcare / Psychology	Data, Design
165	Straw et al., 2022	journalArticle	Healthcare	Data, Model
166	Sun et al., 2020	journalArticle	Computer Science	Data
167	Tasci et al., 2022	journalArticle	Healthcare	Data, Design
168	Tay et al., 2022	journalArticle	Psychology	Data, Design, HAI
169	Timmons et al., 2023	journalArticle	Psychology	Data, Design
170	Ulnicane et al., 2023	journalArticle	Politics / Public Policy	Data, Design, HAI
171	Varona et al., 2022	journalArticle	Information Systems	Data, Design
172	Vokinger et al., 2021	journalArticle	Healthcare	Data, Design
173	Vorisek et al., 2023	journalArticle	Healthcare	Data, Design
174	Walsh et al., 2020	journalArticle	Healthcare	Data, Design
175	Wan et al., 2023	conferencePaper	Social Science	Data, Design
176	Wang et al., 2023	journalArticle	Healthcare	Data, Design
177	Wang et al., 2023	conferencePaper	Social Media	Data, Design, HAI
178	Weber, 2019	journalArticle	Computer Engineering	Data, Design, HAI
179	Wolfe et al., 2023	conferencePaper	Social Science	Data, Design, HAI
180	Yang et al., 2023	journalArticle	Healthcare	Data, Design
181	Yang et al., 2023	journalArticle	Healthcare	Data, Design
182	Yang et al., 2023	journalArticle	Healthcare	Data, Design
183	Yogarajan et al., 2022	journalArticle	Healthcare	Data, Design
184	Yoon et al., 2021	conferencePaper	Healthcare	Data, Design
185	Yu et al., 2020	journalArticle	Healthcare	Data, Design
186	Yuan et al., 2023	conferencePaper	Social Science	HAI, Data, Design
187	Zajko, 2021	journalArticle	Social Science	Data, Design, HAI
188	Zampieri et al., 2023	journalArticle	Atmospheric Science	Data, Design
189	Zhang, 2022	journalArticle	Social Science / Policy	Data
190	Zhang et al., 2023	journalArticle	Public Policy	Data
191	Zhang et al., 2022	journalArticle	Healthcare	Data, Design

#	Author References	Type	In Domain	Bias_Cat- egory
192	Zhang et al., 2022	journalArticle	Physics / Math / Computer Science	Data, Design
193	Zhang et al., 2023	journalArticle	Business	HAI
194	Akter et al., 2021	journalArticle	Information Management	Data, Design
195	Alfarsi et al., 2020	conferencePaper	Education	HAI
196	Bashir et al., 2020	preprint	Computer Science	Design
197	Belkin et al., 2019	journalArticle	Computer Science	Design
198	Bellamy et al., 2018	preprint	Computer Science	Design
199	Bennett et al., 2020	journalArticle	Social Science	Data, Design
200	Boivin et al., 2020	journalArticle	Criminology	Data
#	Author References	Type	In Domain	Bias_Cat- egory
201	Boivin et al., 2017	journalArticle	Criminology	Data
202	Celis et al., 2020	journalArticle	Computer Science	Data
203	HAI Teaming, 2022	book	General	HAI
204	Crockett et al., 2023	conference-Paper	Computer Science	HAI
205	Delgado-Rodriguez, 2004	journalArticle	Public Health	Data
206	Dwivedi et al., 2023	journalArticle	General	Data, Design, HAI
207	Engstrom et al., 2020	journalArticle	Computer Science	Data
208	Friesen et al., 2022	journalArticle	Aerospace Science	HAI
209	Fu et al., 2020	journalArticle	Business	Data, Design
210	Gall et al., 2015	book	Computer Science	Data
211	Garcia et al., 2016	journalArticle	Computer Science	Data
212	Gianfrancesco et al., 2018	journalArticle	Healthcare	Data
213	Gonzalez-Zelaya et al., 2019	conference-Paper	Computer Science	Data
214	Grammenos et al., 2018	journalArticle	Computer Science	Data
215	Grise et al., 2017	journalArticle	Engineering	Data
216	Haibo He; Garcia et al., 2009	journalArticle	Computer Science	Data
217	Hall, 1999	journalArticle	Computer Science	Data

#	Author References	Type	In Domain	Bias_Cat-egory	#	Author References	Type	In Domain	Bias_Cat-egory
218	He et al., 2023	conference-Paper	Computer Science	HAI	244	Rai, 2020	journalArticle	Business	Design
219	Hengstler et al., 2016	journalArticle	Social Science	HAI	245	Raudys et al., 1990	conference-Paper	Math / Statistics	Data
220	Ho et al., 2020	preprint	General	Data	246	Roselli et al., 2019	conference-Paper	Computer Science	Data
221	Houkj et al., 2006	journalArticle	Computer Science	Data	247	Salminen et al., 2019	conference-Paper	Computer Science	Data
222	Howard et al., 2018	journalArticle	Public Policy	Data, Design	248	Samad Dahri et al., 2019	journalArticle	Healthcare	Data
223	Huang et al., 2020	conference-Paper	Computer Science	Design, HAI	249	Samimi et al., 2013	journalArticle	Healthcare	Data
224	Hube et al., 2019	conference-Paper	Computer Science	Data	250	Sevetlidis et al., 2022	journalArticle	Computer Science	Data
225	IBM, 2021	webpage	Computer Science	Design	#	Author References	Type	In Domain	Bias_Cat-egory
226	Jain et al., 2023	conference-Paper	Computer Science	Data, Design	251	Srinivasan et al., 2021	preprint	Computer Science	Design
227	Kalita, 2023	book	Computer Science	Design	252	Srivastava et al., 2018	conferencePaper	Computer Science	Data
228	Kalle et al., 2019	journalArticle	Criminology	Data	253	Struminskaya et al., 2021	journalArticle	Public Opinion	Data
229	Kamiran et al., 2012	journalArticle	General	Data	254	Sun et al., 2009	journalArticle	Computer Science	Data
230	Kammoun et al., 2022	journalArticle	Computer Science	Design	255	Tommasi et al., 2017	journalArticle	Computer Science	Data
231	Kartal, 2022	journalArticle	Economics	Data, Design	256	Torralba et al., 2011	conferencePaper	Computer Science	Data
232	Khetani et al., 2023	journalArticle	Computer Science	Design	257	Torronen, 2001	journalArticle	Social Science	Data
233	Leavy et al., 2020	preprint	General	Data	258	van Stein et al., 2023	preprint	Computer Science	Design
234	Lehner et al., 2009	journalArticle	Systems Engineering	Data	259	Vermetten et al., 2022	journalArticle	Computer Science	Design
235	Lloyd et al., 2018	journalArticle	General	Data	260	Wang et al., 2023	journalArticle	Healthcare	HAI
236	Locke et al., 2024	journalArticle	Business	Design	261	Yigitcanlar et al., 2022	journalArticle	General	Data
237	Lupez et al., 2013	journalArticle	Information Science	Data, Design	262	Zhang et al., 2017	journalArticle	Information Sciences	Data
238	Makhortykh et al., 2021	bookSection	General	Data	263	Zhao et al., 2020	journalArticle	Healthcare	Data, Design
239	Mishraky et al., 2022	journalArticle	Information Technology	Data, Design					
240	Ntoutsis et al., 2020	journalArticle	Computer/Social Science	Data, Design					
241	Olteanu et al., 2019	journalArticle	Information and Communication Technologies	Data					
242	Otterbacher et al., 2019	journalArticle	Computer Science	Data					
243	Pflanzer et al., 2023	journalArticle	General	HAI					

Appendix B

Appendix B - The Network of Sources of Bias in AI-Enabled Systems



Data availability The data supporting this study are derived from public sources, which are cited in the reference section.

Declarations

Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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